

Simple Water Shader

Documentation — for Unity URP | Houdisoft Technology

Overview

Simple Water Shader is a lightweight water shader for Unity's Universal Render Pipeline (URP). It includes animated waves, depth-based shallow/deep water color, smooth transparency, fresnel reflections, animated ripple normals, and shoreline foam — all in a single hand-written shader with no extra render passes.

A one-click menu command is included to add a ready-to-use water object to your scene.

Requirements

- Unity with the Universal Render Pipeline (URP) installed and active.
- Unity 2021.2 / URP 12 or newer is recommended.

Quick Setup

1. Open your URP Asset (Project Settings > Graphics, or Quality > the active render pipeline asset) and enable **Depth Texture** under the Rendering section. This is the only project setting the shader needs.
2. In the menu bar, go to **GameObject > 3D Object > Add Simple Water**. This drops a ready-to-go water plane into your scene, positioned at your Scene view camera.
3. Select the new water object and adjust the material properties in the Inspector to fit your scene (see Material Properties below).

Each water object created this way gets its own material, copied from a shared template ("Water Material Sample", stored in a Resources folder). Editing one water object's properties never affects any other water object.

Material Properties

Waves

Property	What it does
Wave Speed	How fast the waves move.
Wave Strength	How tall the waves are.
Wave Scale	Size of the waves. Higher values give smaller, tighter waves.

Water Color

Property	What it does
Shallow Color	Color used near the shoreline / shallow water.
Deep Color	Color used in deep water.
Water Depth	How many world units deep the water needs to be before it reads as fully “deep” colored. Raise this for a more gradual color fade across a large lake; lower it for a tighter shoreline gradient.

Ripples

Property	What it does
Normal Map	A tileable normal map used for small surface ripples.
Normal Tiling	How many times the normal map repeats across the surface.
Normal Strength	How pronounced the ripples look.
Normal Speed	How fast the ripples scroll across the surface.

Reflection

Property	What it does
Fresnel Power	Controls how the reflection intensity changes with viewing angle — higher values concentrate reflections at glancing angles.
Reflection Strength	Overall intensity of the reflection.

Reflections are sampled from the nearest reflection probe (or the skybox if none is present). Add a Reflection Probe to your scene for the best result.

Foam

Property	What it does
Foam Color	Color of the shoreline / underwater-object foam.
Foam Distance	How far, in world units, the foam extends from the shoreline or a submerged object.
Foam Noise Texture	A grayscale noise texture that breaks up the foam edge so it doesn’t look like a hard line.
Foam Tiling	Tiling of the foam noise texture.
Foam Speed	How fast the foam noise scrolls.

How Depth Detection Works

The shader reads the camera's depth texture to find whatever is behind the water at each pixel (the lake bed, a submerged rock, and so on), reconstructs that point's actual position in the world, and compares its height to the water surface's own height. The vertical gap between them is used as the water depth.

This keeps the shallow/deep gradient and foam accurate from any camera angle, and works on a plain flat plane — no special mesh shaping is required.

Tips for Best Results

- Use a Plane with a reasonable number of subdivisions if you want smoother-looking wave motion at larger scales — a coarse mesh will show visible facets as it moves.
- A soft, low-contrast noise texture (clouds/Perlin style) works better for Foam Noise Texture than a sharp or high-frequency one.
- Keep Wave Strength modest (under about 0.3) for calm lakes/ponds; raise it along with Wave Scale for choppier open water.

Troubleshooting

Water is one flat color, with no shallow/deep gradient or foam

Enable **Depth Texture** in your URP Asset (see Quick Setup, step 1).

Water renders bright pink/magenta

This means the shader failed to compile. Open the Console window for the exact error and check the Support section below — this is almost always caused by a Unity/URP version difference in an internal include path.

No reflections are visible

Add a Reflection Probe to your scene, or make sure a skybox is assigned (Window > Rendering > Lighting).

Foam isn't showing up

Confirm Foam Noise Texture is assigned and Foam Distance isn't set too low, and double-check Depth Texture is enabled.

The GameObject > 3D Object > Add Simple Water menu item is missing

Make sure the package's Editor script has imported correctly and sits inside a folder named exactly "Editor" somewhere under Assets.

Support

Questions or issues? Reach out any time:

- Email: ahmedhouididev@gmail.com

- Publisher page: assetstore.unity.com/publishers/105962

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